Workflow Setup

To use this workflow, the following steps are required. **Import Workflow:**

1. Open n8n dashboard
2. Click **Import Workflow**
3. Upload the provided JSON workflow file
4. The workflow nodes will appear in the editor.

Credentials Configuration

After importing the workflow, the following credentials need to be configured:

Telegram Credential → Add bot token
Google Sheets Credential → Connect Google account
Gmail Credential → Connect Gmail account
Groq API Key → Add in HTTP request node

Running the System

To start the system:

1. Open the workflow in n8n
2. Verify all credentials are connected
3. Click **Activate Workflow**

Now the automation system will start working.

Telegram-Based AI Service Booking System

Project Overview
In this project, I built an **AI-powered service booking system** using **Telegram Bot, n8n, Groq AI, Google Sheets, and Gmail**.
Users can interact with a **Telegram chatbot** to ask about available products or services. The bot reads product data from **Google Sheets**, processes the user's request using **Groq AI**, and responds accordingly.
When a user makes a booking request, the system **stores the booking in Google Sheets** and **sends a confirmation email automatically**.

Overall, the system **automates service inquiries, bookings, and confirmations**.

System Requirements

To run this project, the following services are required:

• n8n account (Cloud or self-hosted)
• Telegram account
• Telegram Bot Token
• Google account (for Google Sheets)
• Gmail account (for sending confirmation emails)
• Groq API Key (for AI conversation)

System Setup

First, I created an n8n workflow **which connects Telegram, AI processing, Google Sheets, and Gmail together. The system works like this:**

Customer sends a message in Telegram →
Telegram Trigger receives the message →
The workflow reads available products from Google Sheets →
Groq AI processes the message →
The chatbot responds to the user →
If the user confirms booking →
Booking information is saved in Google Sheets →
Confirmation email is sent to the customer.

Telegram Bot Configuration

To create the Telegram bot, I used BotFather**.**

Steps I followed:

1. Open Telegram and search **BotFather**
2. Start the chat and type "/newbot"
3. Enter the bot name and username.

After that, BotFather generated an **API Token** for the bot. This token is required to connect the Telegram bot with **n8n**. Inside n8n, I added the **Telegram Trigger node** and pasted the bot token in the credentials section. Then I set the trigger event to: "Message"

Now whenever a user sends a message to the bot, the workflow starts automatically.

Google Sheets Configuration

In this project, I used Google Sheets as the database**. Two sheets are required.**

Product Sheet

This sheet stores all product or service information provided by the shop owner. Columns I used:

| Product Name | Description | Price | Availability | Service Duration |

Inside n8n, I added a **Google Sheets node** and selected the **Read operation** to fetch product information from the sheet.

Booking Sheet

Another sheet is used to store booking data. columns:

| Customer Name | Customer Query | Date and time|

Whenever a user requests a service, the system saves the booking information in this sheet automatically.

Groq AI Configuration

For AI conversation, I used **Groq AI**. Steps I followed:

- Created an account and Generated a **Groq API Key**

Inside n8n, I added an **HTTP Request node** and connected it to the Groq API. “API Endpoint used and Model used”

The user's message from Telegram is sent to Groq AI, and the AI generates a response which is then sent back to the user through Telegram.

Email Confirmation Setup

To send booking confirmation emails, I used the **Gmail node in n8n**.

Steps: First Add **Gmail Node** then Select operation. For Send Message

- First Connected Gmail account and then Configure the email template.

Email message:
Subject: Booking Confirmation
Hello,
Your booking request has been received successfully.
Our team will contact you shortly to confirm your service.
Thank you.

Whenever a booking request is recorded, this email is automatically sent to the customer

N.B.: Due to the trial version of n8n, I faced some limitations and could not fully resolve all the issues. However, I want to be honest about this. Since I have experience working on production-level projects, I am confident that I can deliver my best and contribute to the growth of the company.